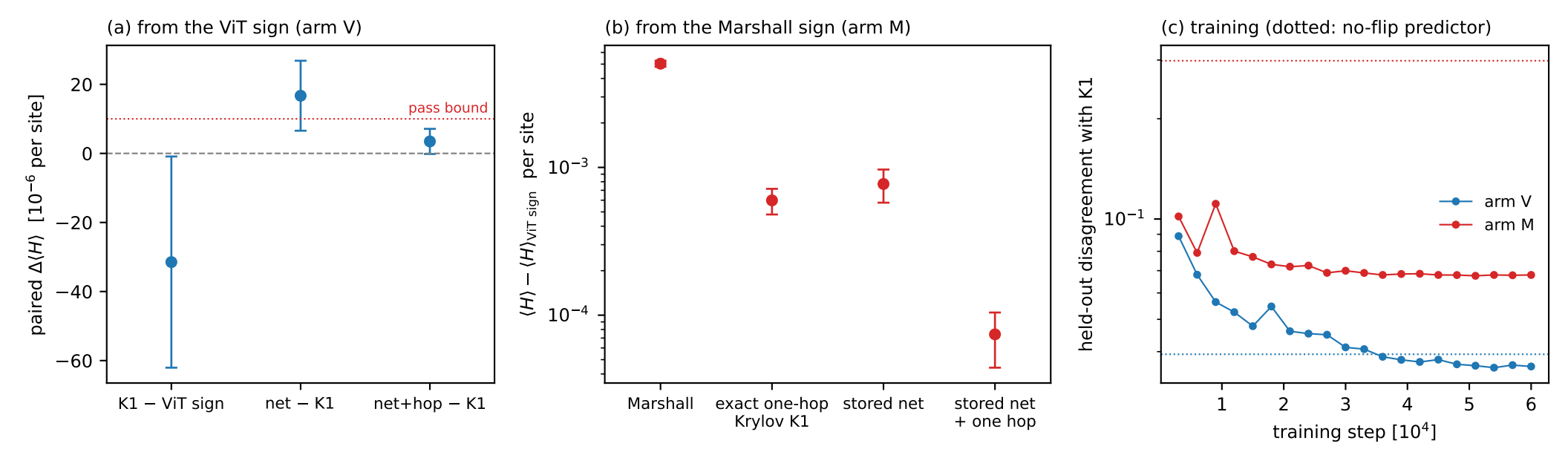
(a) from the ViT sign (arm V)
paired Δ⟨H⟩ [10⁻⁶ per site]
20
0
−20
−40
−60
pass bound
K1 − ViT sign
net − K1
net+hop − K1
(b) from the Marshall sign (arm M)
⟨H⟩ − ⟨H⟩ViT sign per site
10⁻³
10⁻⁴
Marshall
exact one-hop Krylov K1
stored net
stored net + one hop
(c) training (dotted: no-flip predictor)
held-out disagreement with K1
10⁻¹
arm V
arm M
1
2
3
4
5
6
training step [10⁴]